

K-FIELD: CMB-PUMPED ARMATURE ANALYSIS

Diametric Coil Pairs, Rotating Phase Harmonics, Electrical Frequency, and Staged Tap Sequence

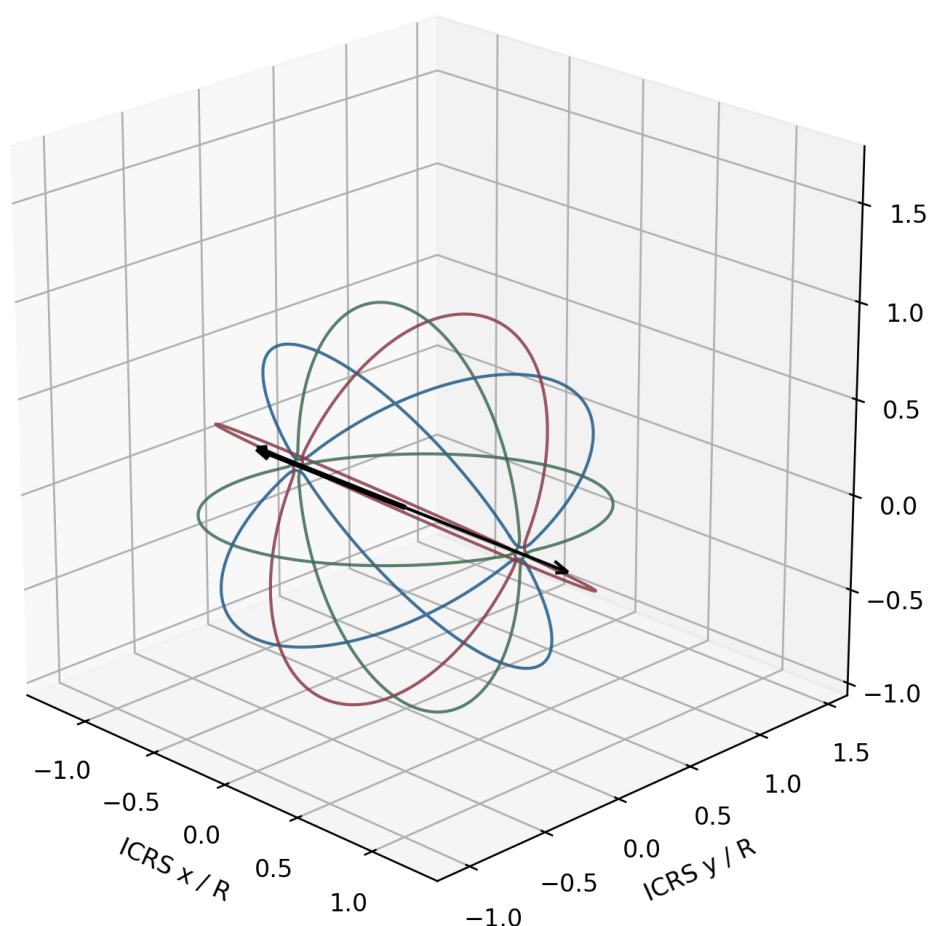
P. Dwayne Esterline, Principal Investigator

BS, Huntington University (1999) | MBA, Indiana Wesleyan University (2008)

Krytronx

Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.

Three staged diametric pairs: six complementary closed coils



Title figure. Three equidistant diametric coil pairs are generated by rigidly rotating the computed closed Earth-CMB wire route about its common apex-antapex axis. The six centerlines are dimensionally correct on the normalized route sphere $R=1$. The rotation axis is the authoritative Earth-CMB vector in ICRS coordinates.

Archival foundational education reference - Unix epoch 1784661365

Abstract

This paper develops the rotating-armature extension of the closed Earth-CMB wire route. The base wire follows two distinct meridian trajectories joined by finite-radius sinusoidal-curvature bridges. In the fixed K-Field orientation, forward and reverse circulation complete the same closed-loop scalar cost but accumulate that cost in different orders. The half-loop cumulative difference provides a signed, rotation-dependent armature signal $H(\phi)$. Rigid rotation about the Earth-CMB axis converts the static spatial anisotropy into a periodic waveform. Numerical rotation of the complete smoothed centerline gives the dominant fit $H(\phi) = 2.706182128080 \cos(2\phi + 0.065460604099 \text{ deg})$. The response therefore has two electrical cycles per mechanical revolution, equivalent to two pole pairs or a four-pole spatial field. A diametric mate rotated by 180 mechanical degrees has the same principal electrical phase because the dominant $k=2$ harmonic advances by 360 electrical degrees. Same-polarity addition of the two coils reinforces the even channel to 5.412364256160 at the fundamental and suppresses odd imperfections. Opposite-polarity subtraction removes the principal even channel and isolates a once-per-revolution index signal with amplitude 0.029485525211 and phase 59.579831538 deg. True opposition of the principal channel occurs at 90 mechanical degrees. For P equidistant diametric pairs, pair axes are separated by $180/P$ mechanical degrees and their principal phases by $360/P$ electrical degrees. The electrical frequency is twice shaft frequency, and the staged tap-event rate is $2P$ times shaft frequency. Positive rotation about the CMB axis selects pair indices $0, P-1, P-2, \dots, 1, 0$; negative rotation selects $0, 1, 2, \dots, P-1, 0$. A three-pair implementation has axes at 0, 60, 120 mechanical degrees, electrical phases 0, 120, 240 degrees, and the positive sequence A-C-B-A. The report distinguishes the exact geometric phase and tap law from voltage, torque, and power, which require an additional constitutive device coupling.

Central result

The closed-route hysteresis signal behaves as a two-pole-pair rotating field. One mechanical revolution produces two principal cycles. A 180 degree mate reinforces the principal channel; a 90 degree rotation opposes it. Equidistant pair ensembles form a balanced polyphase set and must be staged, rectified, or phase processed rather than directly summed.

Contents

Abstract	2
1. Scope and Coordinate Conventions	4
2. Authoritative Cell3 Base Geometry and Plane Systems	4
3. Quadratic Phase Metric and Earth-CMB Alignment	6
4. Closed Wire Route and Base Cumulative Hysteresis	7
5. Rotating Armature Signal	9
6. Principal Phase and Frequency	10
7. Diametric Complementary Coil Pair	11
8. Equidistant Ensemble of P Diametric Pairs	12
9. Correct Tap Sequence Under Rotation	13
10. Three-Pair Armature Example	14
11. Frequency and Shaft-Speed Scaling	16
12. Polyphase Drive, Commutation, and Tap Polarity	17
13. Physical Interpretation and Constitutive Boundary	18
14. Reproducibility and Numerical Precision	18
15. Foundational Consequences	19
Appendix A. Complete Cell3 Half-Step Node Registry	20
Appendix B. Cell3 Vertices and Six Support Planes	21
Appendix C. Rotational Fourier Coefficients	21
Appendix D. Tap Sequences for P=1 Through P=12	22
Appendix E. Machine-Readable Companion	22
Archival Colophon	23

1. Scope and Coordinate Conventions

The analysis uses one fixed ICRS Cartesian orientation and two separate scale domains. The wire route lies on a normalized sphere of radius $R=1$. Cell3 is a distinct native lattice cell expressed in the same ICRS axes. The shared frame allows directions, planes, and rotations to be compared without equating the route radius to a Cell3 length.

The Earth-CMB unit vector points from the antapex toward the apex. Mechanical armature angle ϕ is positive by the right-hand rule about this vector. The source route is $\phi=0$. Every complementary coil is a rigid rotation of the same computed centerline; no RA/Dec interpolation or refitting is introduced.

PRIMARY DEFINITIONS

Earth-CMB axis: $\mathbf{c_hat} = (-0.9707708558359351, 0.2073480091109861, -0.1208749294819919)$

Rotated centerline: $\mathbf{n_phi}(s) = \text{Rot}(\mathbf{c_hat}, \phi) \mathbf{n_0}(s)$

Local scalar density: $q_phi(s) = \sqrt{\mathbf{n_phi}(s)^T \mathbf{G_phase} \mathbf{n_phi}(s)}$

Armature signal: $H(\phi) = C_F(L/2; \phi) - C_R(L/2; \phi)$

2. Authoritative Cell3 Base Geometry and Plane Systems

The base cell provides the native ICRS orientation reference associated with the K-Field. Its three primitive vectors generate eight boundary vertices, twenty-seven centered half-step nodes, three pairs of opposite face planes, and the direct volume. All quantities in this section remain in native Cell3 units.

Authoritative Cell3: 8 vertices, 27 half-step nodes, and 6 face planes

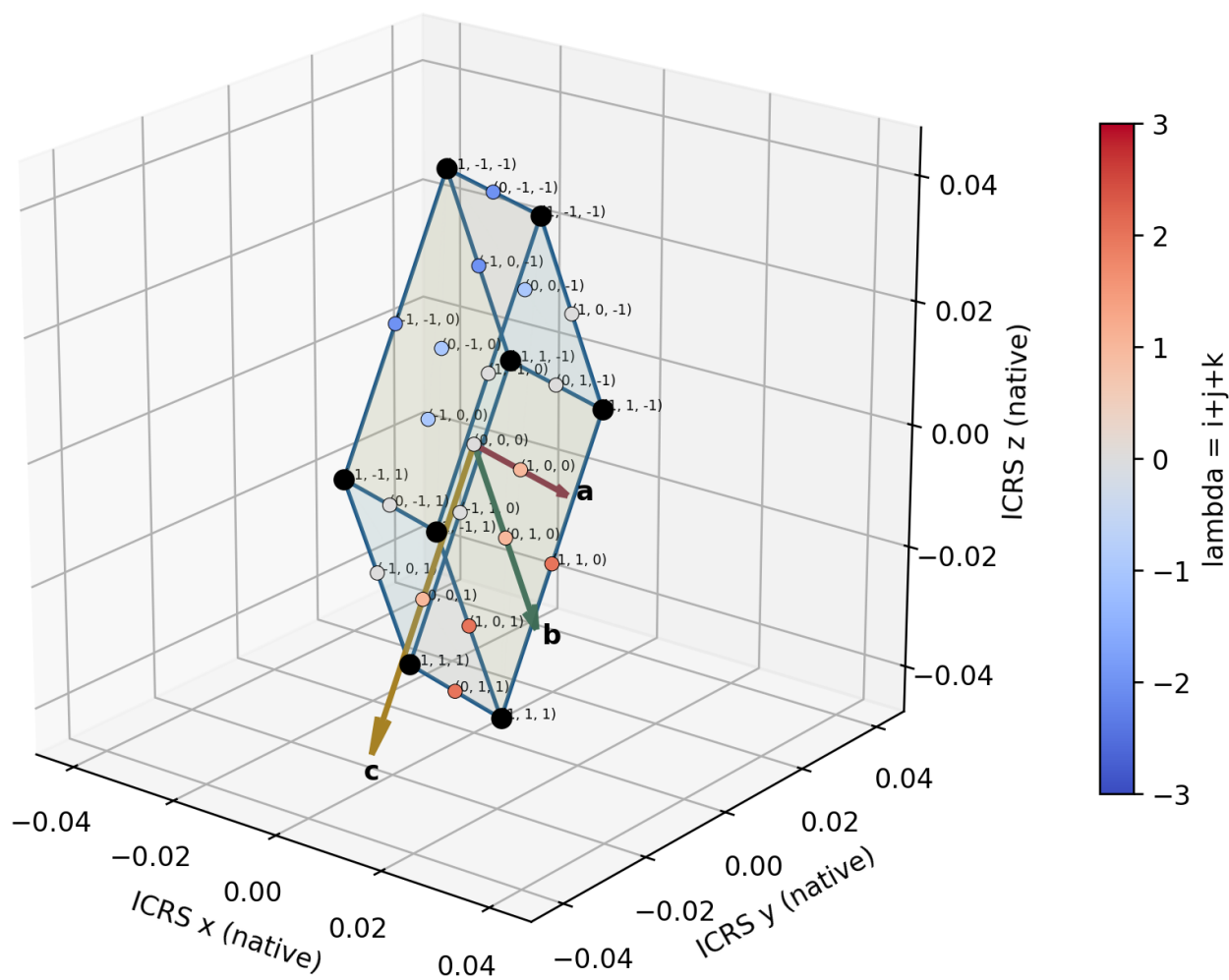


Figure 1. Dimensionally correct authoritative Cell3. All 27 centered half-step nodes are labeled by integer indices (i,j,k), where the stored coordinate is $0.5(i + j + k)$. The six translucent patches are the opposite A_ab, A_ac, and A_bc face planes. The route sphere is not shown because it is a separate scale domain.

vector	ICRS x	ICRS y	ICRS z	length native
a	0.02038143851694645	-0.00341641449641055	-0.00233296591672896	0.02079705878243295
b	0.00219874596832336	0.01290217025869254	-0.03406650153605571	0.03649420513072719
c	-0.00063037948534737	-0.02425772201712787	-0.04292435457109382	0.04930856589985942

face	area native^2	included angle deg	unit normal ICRS	opposite-plane separation native
A_ab	0.00075472178967262	83.933483012313	[0.194092200807768, 0.913177168821229, 0.358379234790804]	0.03765714700038611
A_ac	0.00101125575768807	80.448130018338	[0.089052594101605, 0.866576769291032, -0.491033948326276]	0.02810433380678284
A_bc	0.00138578443173329	50.363231813756	[-0.995965190308961, 0.083602162507377, -0.0326192905644]	0.02050872323810781

The primitive volume is $2.8420669378096486e-05$ native cubed. Opposite faces are translations of the same primitive parallelogram, so their normals and areas are identical while their support offsets have opposite signs.

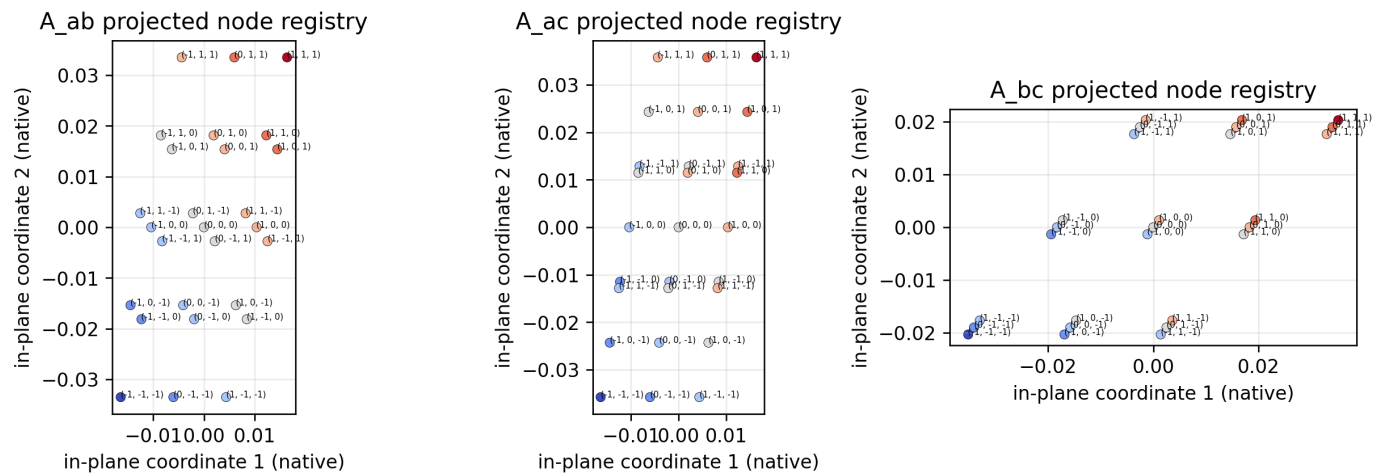


Figure 2. Orthogonal projections of the complete 27-node registry into the three primitive face coordinate systems. The projections are derived from the authoritative basis and preserve native dimensions.

3. Quadratic Phase Metric and Earth-CMB Alignment

The rotating response is evaluated in the supplied symmetric positive-definite quadratic phase metric. At a unit direction $\mathbf{n}$, the scalar cost density is the square root of the quadratic form. Positive definiteness guarantees a real positive density at every route point. Rotating the wire changes its orientation relative to the fixed metric ellipsoid.

row	G _x	G _y	G _z
0	2.9764106378	1.2431507795	-2.0803600354
1	1.2431507795	2.5922286635	-2.3178839937
2	-2.0803600354	-2.3178839937	6.7735709133

property	value
eigenvalues ascending	1.429328753487, 2.112545026447, 8.800336434665
trace	12.342210214600
determinant	26.572803745831
condition number	6.156971524705
CMB apex RA, Dec	167.943299879828, -6.942599922130 deg
CMB antapex RA, Dec	347.943299879828, 6.942599922130 deg

Quadratic phase metric ellipsoid and Earth-CMB rotation axis

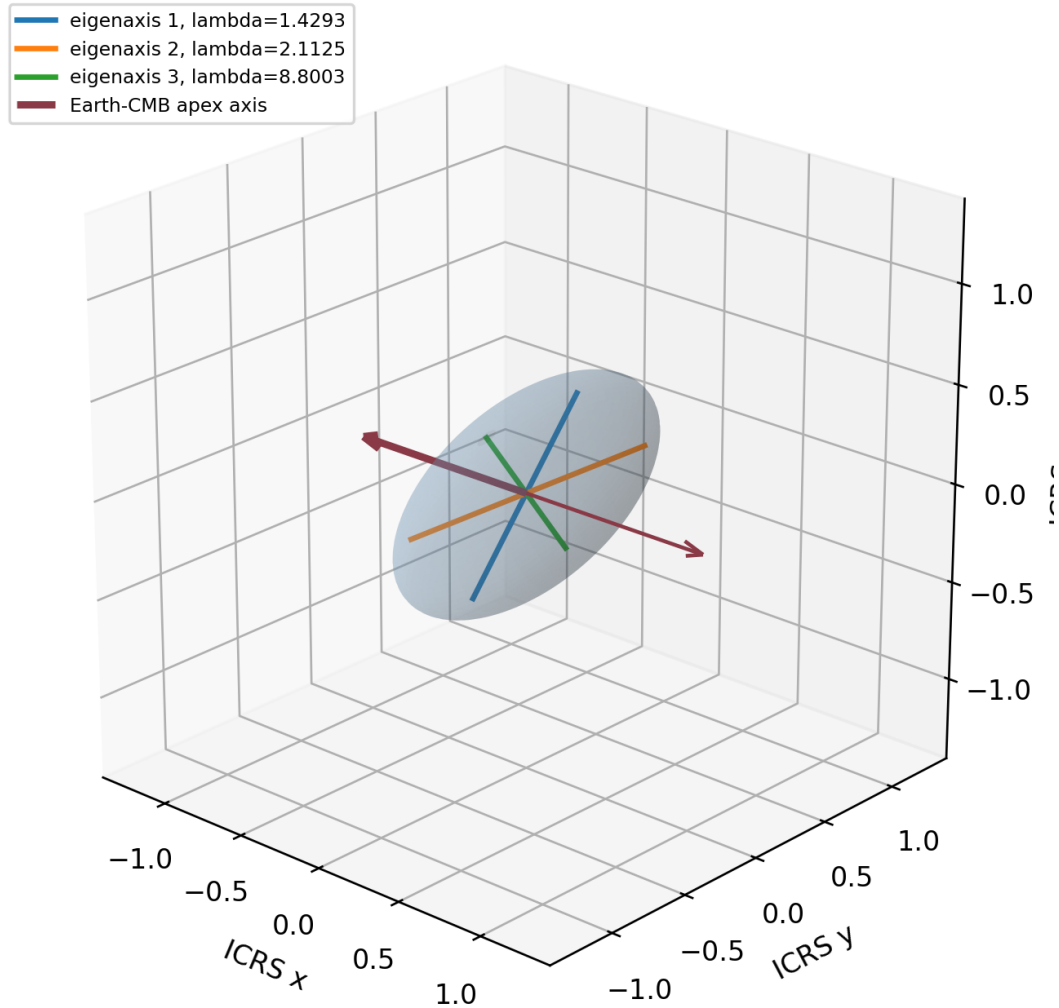


Figure 3. Unit-cost ellipsoid $x^T G_{\text{phase}} x=1$, its three metric eigenaxes, and the Earth-CMB armature axis. Rotation about the CMB axis sweeps the complete route through this anisotropic metric.

4. Closed Wire Route and Base Cumulative Hysteresis

The wire follows the exact smoothed roller-coaster centerline. The principal high-cost meridian runs from CMB antapex to apex. A finite-radius bridge rotates the tangent into the return meridian. The return reaches the antapex and a second bridge closes the wire. The bridges use a sinusoidal geodesic-curvature profile with zero curvature and zero curvature rate at the meridian interfaces.

Single closed wire route aligned to the Earth-CMB axis

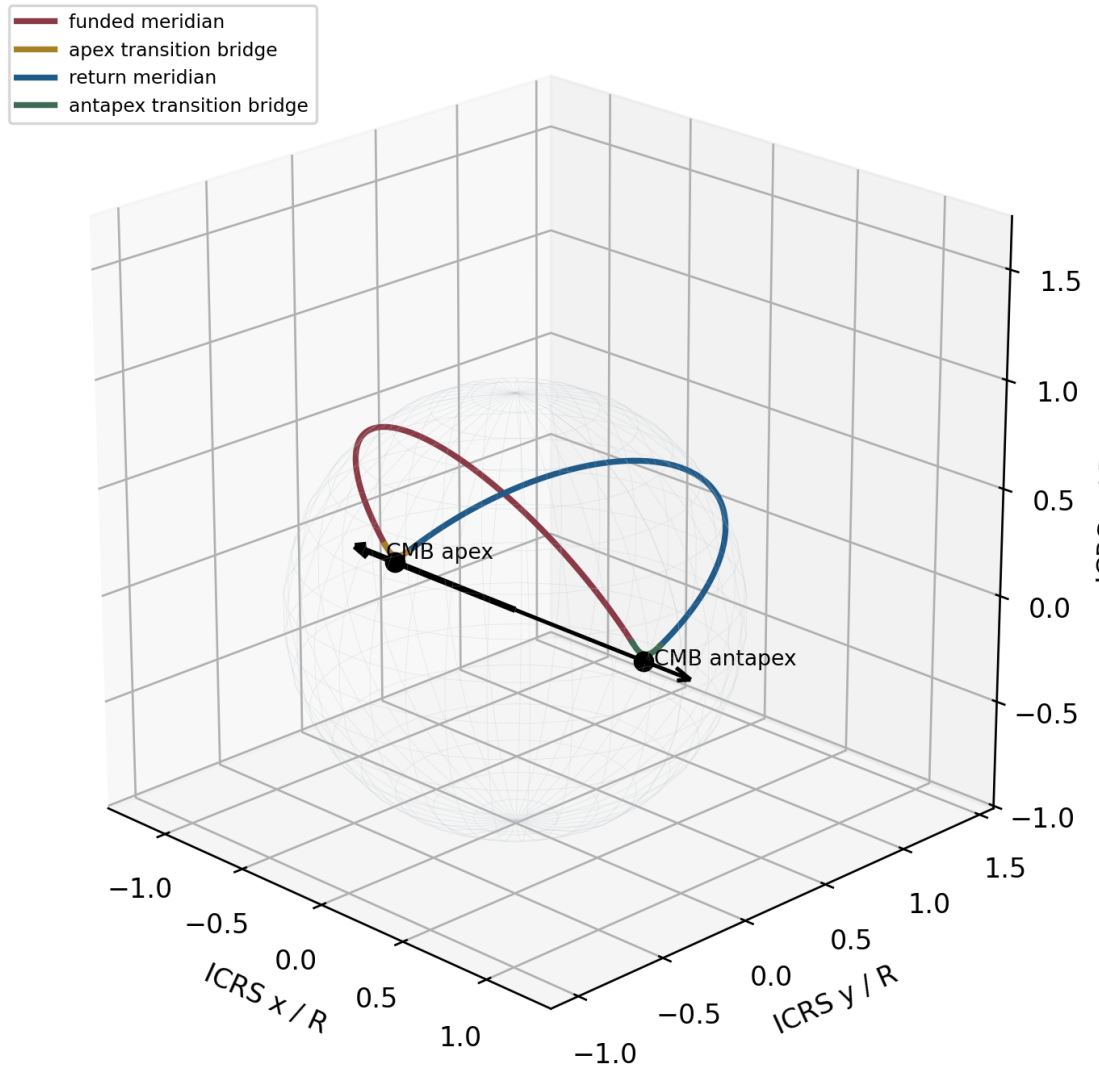


Figure 4. The single-turn wire route on the normalized sphere. The two meridian legs are distinct great-circle paths sharing the Earth-CMB diameter. The compact apex and antapex bridges provide continuous position, tangent, force, and jerk.

quantity	value
closed wire length	6.223020710975859 R
sample intervals	14400
sample spacing	0.000432154216039990 R
meridian interface tangent rotation	89.769450000754 deg
bridge length each	0.179364667368454 R
closed-loop scalar cost at source orientation	11.276251390281310
high-minus-return half-loop signal	2.748220082602268

Forward and reverse circulation finish with the same total scalar cost because they traverse the same closed wire. They do not, however, accumulate cost in the same order. At equal distance traveled, their cumulative traces separate and reunite only at the completed loop.

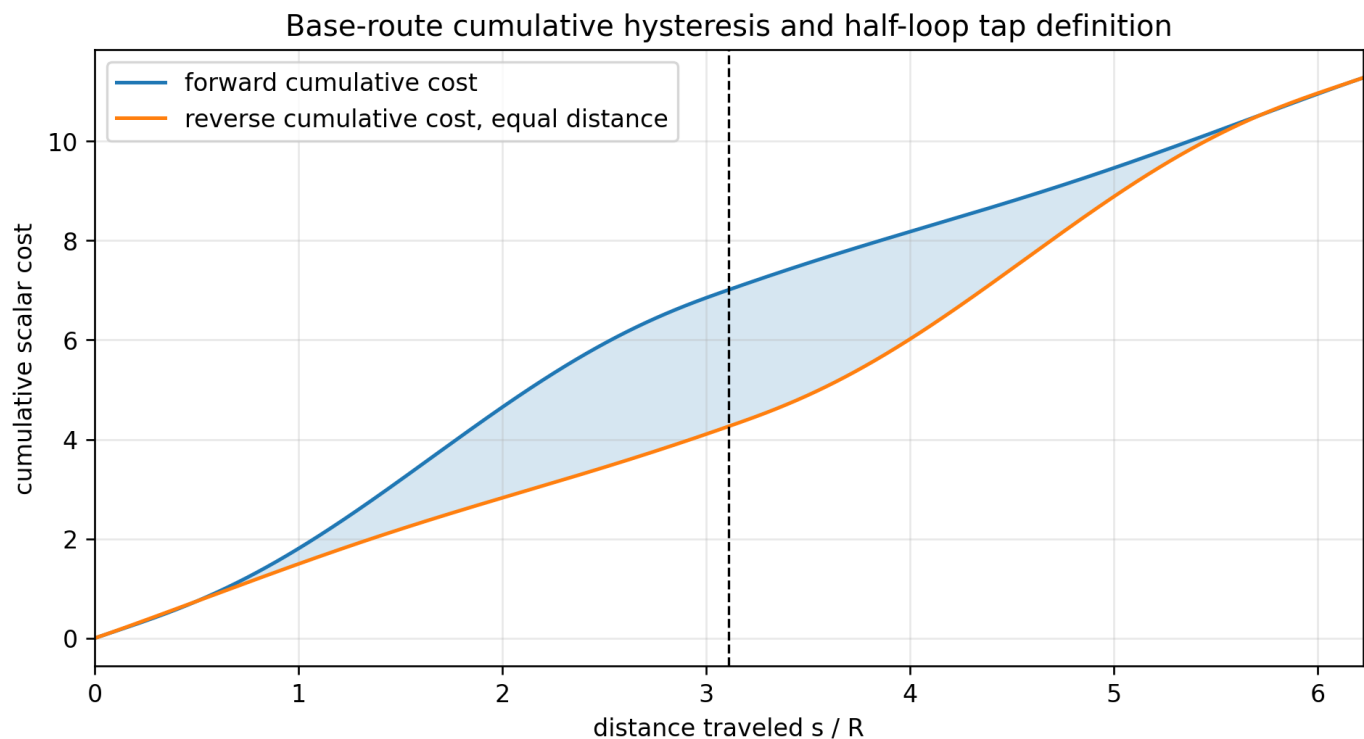


Figure 5. Forward and reverse cumulative cost at the source orientation. The half-loop tap is the maximum separation point: $C_F=7.012235736442$, $C_R=4.264015653840$, and $H=2.748220082602$.

fixed-route hysteresis quantity	value
$C_F(L/2)$	7.012235736441789
$C_R(L/2)$	4.264015653839521
$H(0)=C_F-C_R$ at half loop	2.748220082602268
signed cumulative-loop area	7.596421328301542 cost x R
absolute cumulative-loop area	7.617373149317954 cost x R

5. Rotating Armature Signal

For every mechanical angle ϕ , the complete centerline is rotated rigidly about the Earth-CMB axis and the local scalar density is recomputed. The half-loop cumulative difference is then evaluated using the same geometric start point and the same forward-oriented wire coordinate. This produces a periodic armature waveform without changing the wire shape.

ROTATING SIGNAL

$C_F(L/2;\phi) = \text{integral over the first half of the rotated wire}$

$C_R(L/2;\phi) = \text{integral over the second half encountered from the same start in reverse circulation}$

$H(\phi) = C_F(L/2;\phi) - C_R(L/2;\phi)$

$H(0) = 2.748220082602268$

The computed waveform is dominated by spatial harmonic $k=2$. The small $k=1$ component records the part that changes under a 180 degree rotation. Higher even harmonics arise because the square root of the rotated quadratic form is not itself a finite quadratic polynomial.

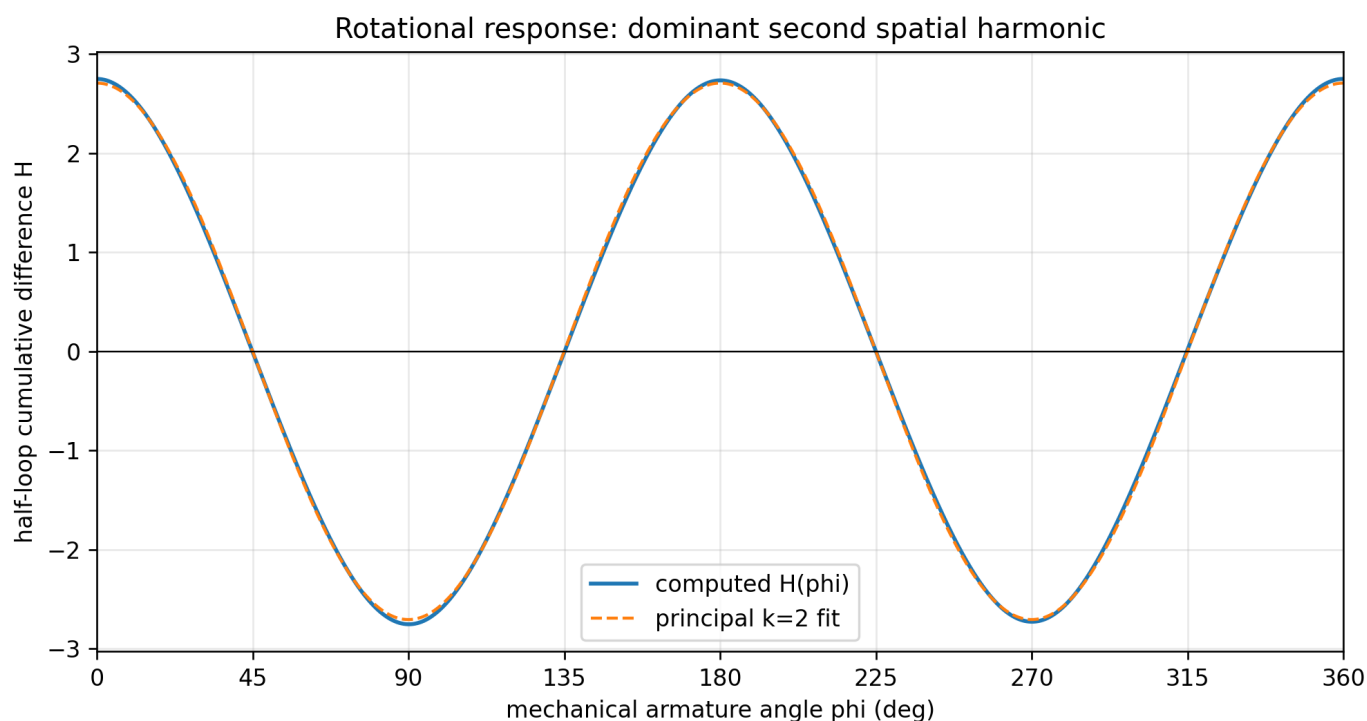


Figure 6. Direct numerical armature sweep compared with the dominant second-harmonic fit. The computed route and both endpoint bridges are rotated; no simplified circular approximation is used.

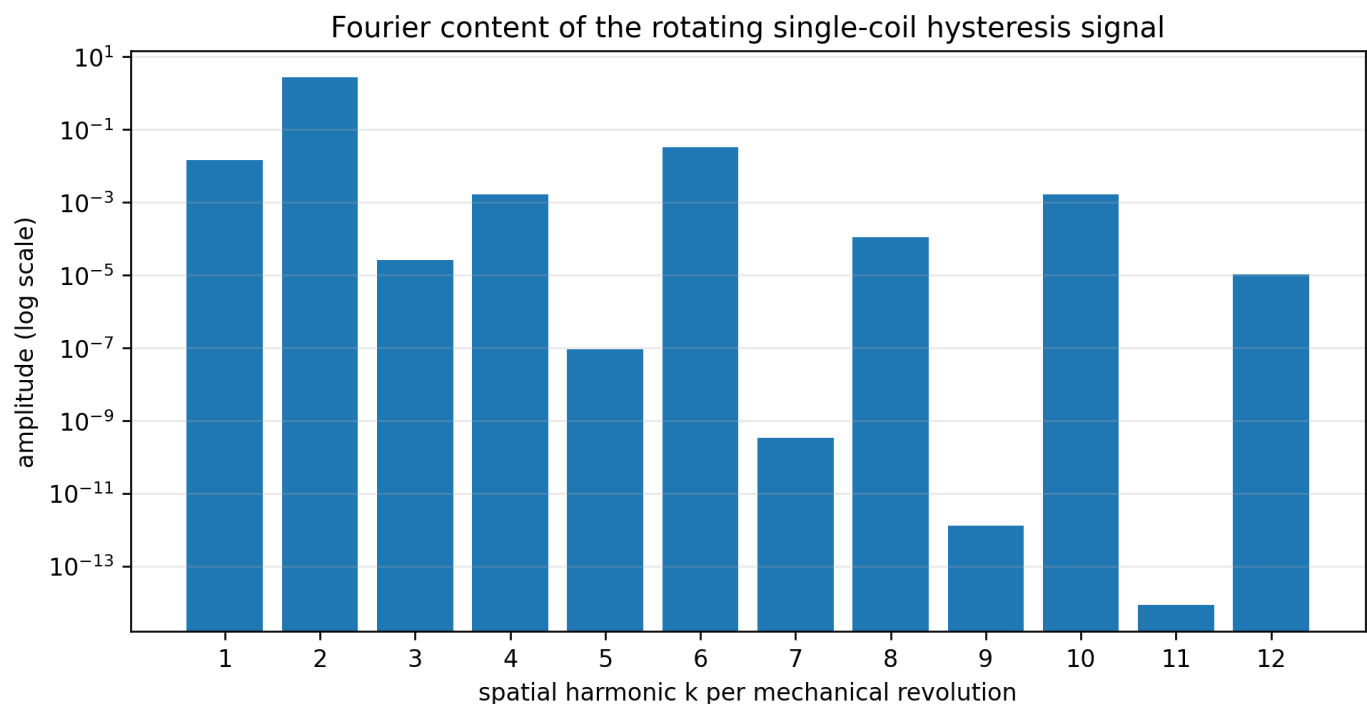


Figure 7. Fourier amplitudes through spatial harmonic $k=12$. The $k=2$ term dominates. The $k=6$ term is the third electrical harmonic because it completes three cycles for each $k=2$ electrical cycle.

6. Principal Phase and Frequency

The dominant fit to the full numerical sweep is

```

PRINCIPAL ROTATING HARMONIC
H(phi) approximately 2.70618212807980 cos(2 phi + 0.06546060 deg)
theta_e = 2 phi + 0.06546060 deg
omega_e = 2 omega_m
f_e = 2 f_m = N_rpm / 30

```

The coefficient 2 multiplying phi is the pole-pair count of the principal spatial response. The armature therefore behaves as a four-pole field, or two pole pairs. One mechanical revolution produces two complete principal hysteresis cycles.

The principal-fit maximum occurs 0.03273030 mechanical degrees before the defined source zero. Including all resolved harmonics, the actual computed maximum occurs at -0.06063889 degrees with $H=2.748226917095$. The actual minimum occurs at 90.10563784 degrees with $H=-2.753397950960$.

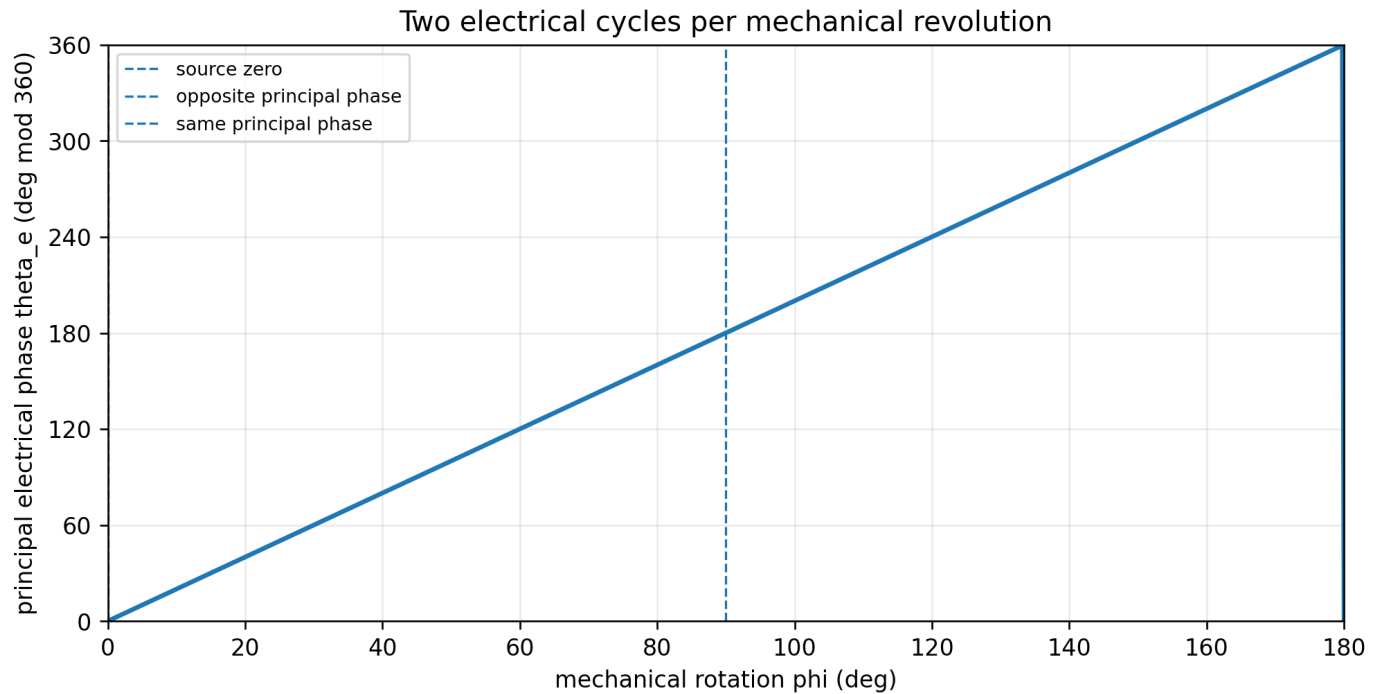


Figure 8. Mechanical-to-electrical phase mapping. A 90 degree mechanical rotation advances the principal phase by 180 electrical degrees; a 180 degree mechanical rotation advances it by 360 electrical degrees.

mechanical rotation	principal electrical change	principal relation
0 deg	0 deg	source phase
45 deg	90 deg	quadrature
90 deg	180 deg	opposite principal phase
180 deg	360 deg	same principal phase
360 deg	720 deg	two complete electrical cycles

7. Diametric Complementary Coil Pair

Let the mate be a rigid copy of the first coil rotated by 180 mechanical degrees about the central axis. Its signed response is $H_{pi}(\phi)=H(\phi+\pi)$. The principal phase advance is 2π , so the two coils are in the same principal electrical phase, not opposite phase.

```

DIAMETRIC PAIR PARITY
H_pi(phi) = H(phi + pi)
Delta theta_e = 2 pi = 360 deg electrical
H_sum(phi) = H(phi) + H(phi + pi)
H_difference(phi) = H(phi) - H(phi + pi)

```

Same-polarity connection preserves the funded-minus-return sign of both coils. It doubles all even spatial harmonics and cancels all odd spatial harmonics. Opposite-polarity connection reverses one mate lead assignment; it cancels the even channel and doubles the odd channel.

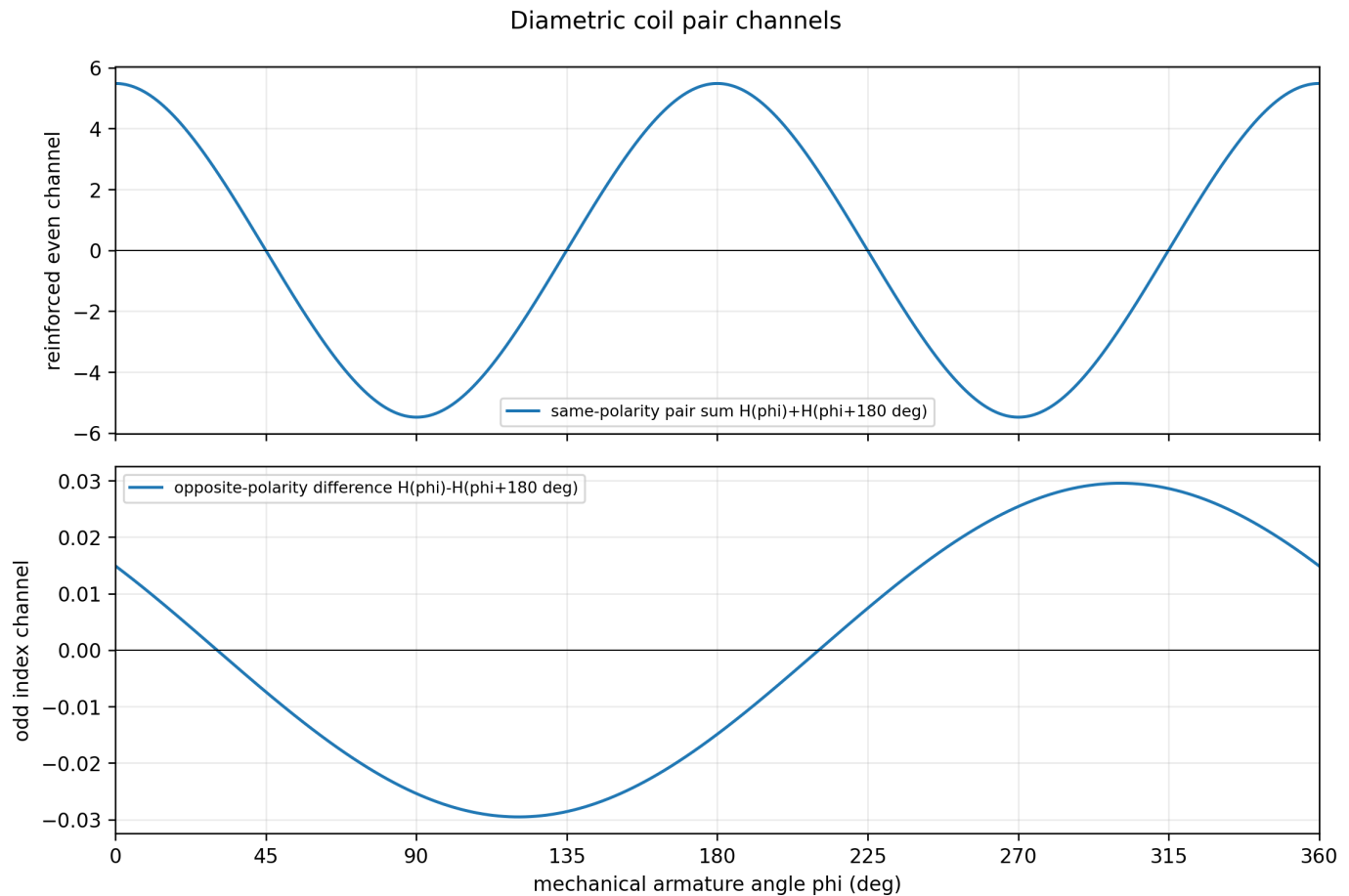


Figure 9. The same-polarity pair sum is the reinforced two-cycle channel. The opposite-polarity difference is a small once-per-revolution index channel.

pair quantity	computed value
principal sum fit	$5.41236425615960 \cos(2 \phi + 0.06546060 \text{ deg})$
actual same-polarity pair maximum	5.481553693621
actual pair maximum angle	-0.00074957 deg
third electrical harmonic amplitude	0.06567179123047
third/fundamental ratio	1.213366 percent
index-channel principal fit	$0.02948552521129 \cos(\phi + 59.57983154 \text{ deg})$
actual index-channel maximum	0.029529456283
actual index maximum angle	300.57569626 deg

A true opposite principal phase occurs at 90 mechanical degrees, not 180 degrees. Therefore, a 180 degree mate should have its leads reversed only when cancellation of the principal channel is intended. For reinforcement, both diametric coils are tapped with the same signed funded-minus-return polarity.

8. Equidistant Ensemble of P Diametric Pairs

Let P denote the number of diametric pair axes. This P is the winding-stage count and must not be confused with the principal pole-pair count, which is fixed at two by the $k=2$ response. Pair axis j is placed at $\alpha_j = j \pi / P$ and its mate at $\alpha_j + \pi$.

```

STAGED ENSEMBLE
alpha_j = j pi / P, j = 0,...,P-1
adjacent pair-axis spacing = 180/P deg mechanical
adjacent principal phase spacing = 360/P deg electrical
S_j(t) approximately A_pair cos(2 omega_m t + 2 pi j/P + delta_2 + beta)
f_tap = 2 P f_m = P N_rpm / 30

```

Here beta is an optional electrical lead or lag angle. Geometry fixes the zero-lead phase and sequence. A torque-optimal or power-optimal beta requires the constitutive relation between cumulative K-Field state and the actual electrical or mechanical device response.

Equal direct summation of all P balanced pair channels cancels the principal harmonic for every P greater than one because the phase vectors are uniformly distributed around the electrical circle. The ensemble must therefore be commutated, rectified, synchronously demodulated, or driven as a polyphase system.

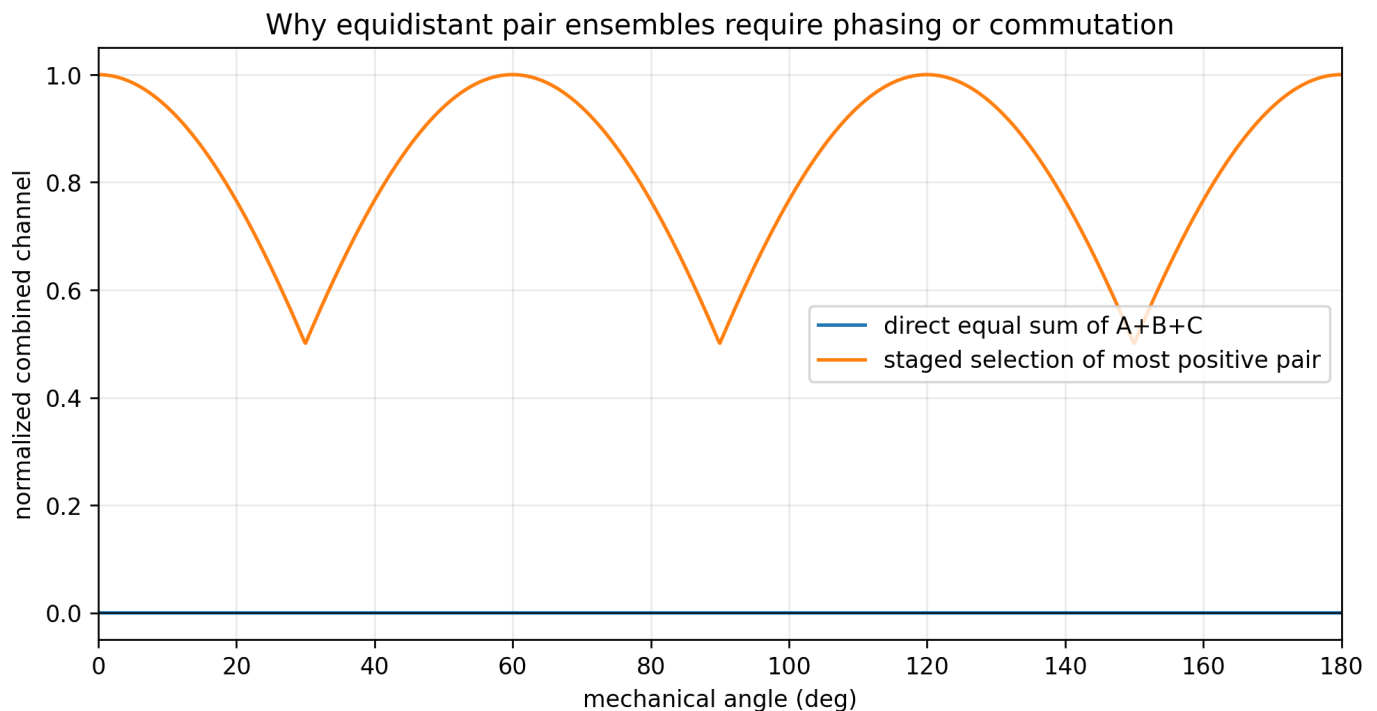


Figure 10. For P=3, direct equal summation of the balanced channels cancels the principal component. Staged selection of the currently positive pair preserves a unidirectional tapped response.

9. Correct Tap Sequence Under Rotation

Label pair axes 0 through P-1 in increasing positive angular order about the positive Earth-CMB axis. Shift the mechanical encoder origin so pair 0 is at its principal positive maximum. For positive shaft rotation, the next positive maximum belongs to the pair at the largest indexed negative phase, which is pair P-1. For negative shaft rotation, the sequence advances through increasing indices.

```

TAP LAW
positive CMB-axis rotation: 0, P-1, P-2, ..., 1, 0
negative CMB-axis rotation: 0, 1, 2, ..., P-1, 0
commutation interval = 180/P deg mechanical
active zero-lead sector width = 180/P deg mechanical
sequence repetitions per mechanical revolution = 2

```

The tap centers occur at $\phi_k = k \pi / P$ relative to the pair-0 maximum. The zero-lead commutation boundaries are halfway between centers, at $\phi = (k+1/2) \pi / P$. Reversing the shaft reverses the pair order. Adding a controlled lead beta translates every boundary by the same mechanical angle $\beta/2$ without changing the order.

P	axis spacing deg	electrical spacing deg	positive rotation order	negative rotation order	events/rev
1	180.000000	360.000000	0-0	0-0	2
2	90.000000	180.000000	0-1-0	0-1-0	4
3	60.000000	120.000000	0-2-1-0	0-1-2-0	6
4	45.000000	90.000000	0-3-2-1-0	0-1-2-3-0	8
5	36.000000	72.000000	0-4-3-2-1-0	0-1-2-3-4-0	10
6	30.000000	60.000000	0-5-4-3-2-1-0	0-1-2-3-4-5-0	12
8	22.500000	45.000000	0-7-6-5-4-3-2-1-0	0-1-2-3-4-5-6-7-0	16
12	15.000000	30.000000	0-11-10-9-8-7-6-5-4-3-2-1-0	0-1-2-3-4-5-6-7-8-9-10-11-0	24

10. Three-Pair Armature Example

The smallest balanced three-phase arrangement uses $P=3$ diametric pairs. Each pair contains two coils separated by 180 mechanical degrees. The pair axes are separated by 60 mechanical degrees, corresponding to 120 electrical degrees.

Three staged diametric pairs: six complementary closed coils

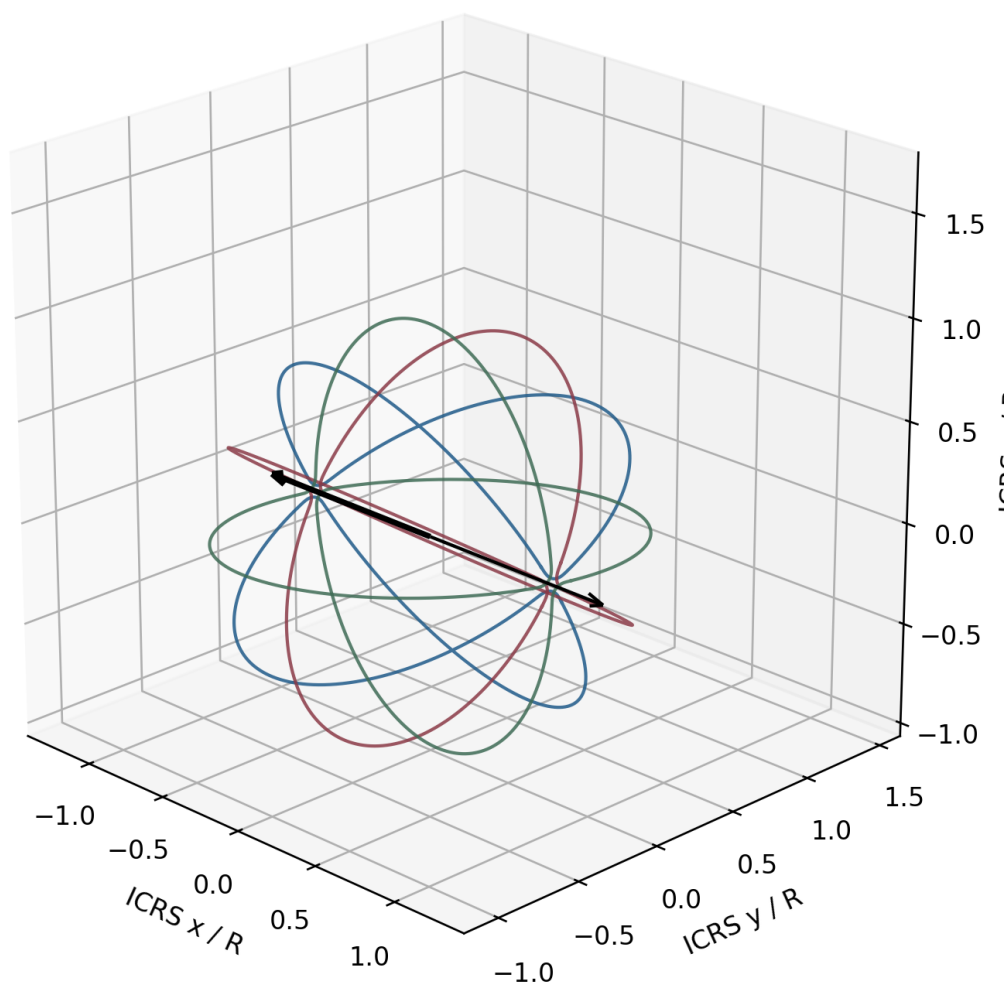


Figure 11. Six physical coil centerlines forming three diametric pairs. Each pair is one electrical phase channel after its two 180 degree mates are combined with the chosen pair polarity.

P=3 pair axes and 120 deg electrical phase separation

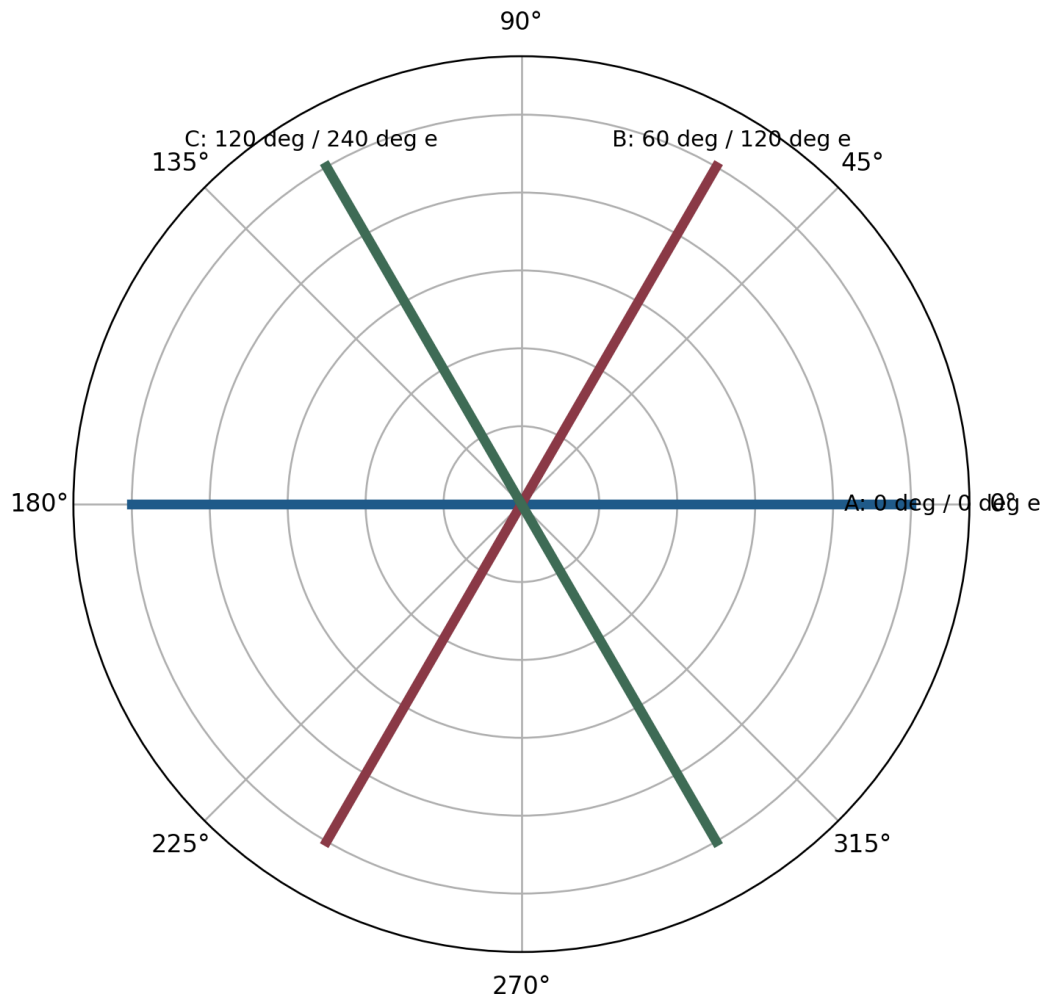


Figure 12. Transverse phase-ring representation of the P=3 geometry. The mechanical pair axes at 0, 60, and 120 degrees correspond to electrical phases 0, 120, and 240 degrees.

pair	mechanical axes	electrical phase
A	0 deg, 180 deg	0 deg
B	60 deg, 240 deg	120 deg
C	120 deg, 300 deg	240 deg

For positive rotation about the Earth-CMB axis, the tap sequence is A-C-B-A. For negative rotation, it is A-B-C-A. Commutation occurs every 60 mechanical degrees and the sequence repeats twice per mechanical revolution.

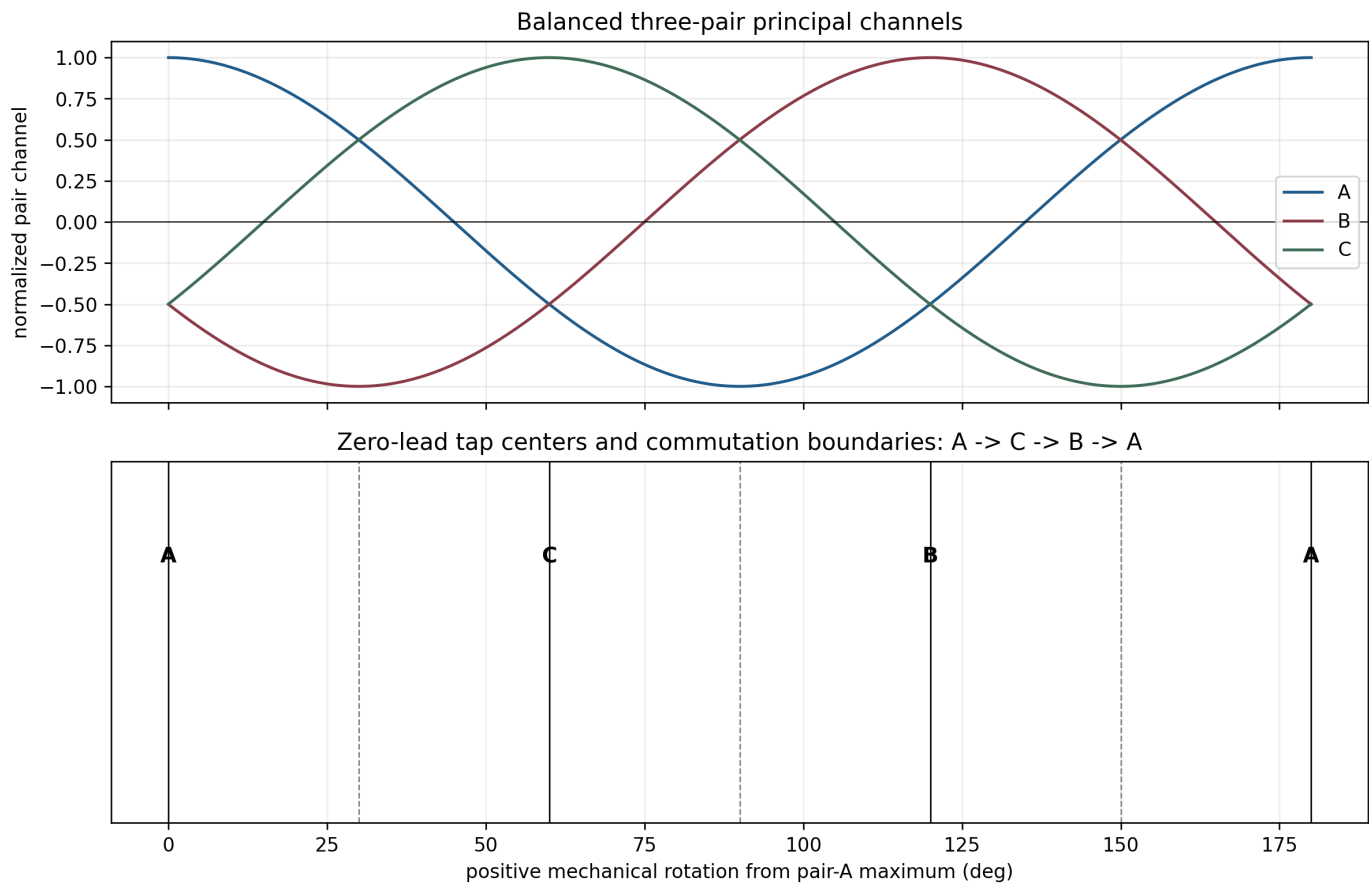


Figure 13. Computed three-pair principal channels and the positive-rotation zero-lead tap sequence. Dashed lines mark the 30, 90, and 150 degree commutation boundaries.

11. Frequency and Shaft-Speed Scaling

Mechanical frequency is shaft revolutions per second. The principal electrical response completes two cycles per revolution. A P-pair commutator has P positive pair maxima per half revolution and therefore 2P tap events per full revolution.

FREQUENCY RELATIONS

$$f_m = N_{rpm} / 60$$

$$f_e = 2 f_m = N_{rpm} / 30$$

$$f_{tap} = P f_e = P N_{rpm} / 30$$

$$\text{time between taps} = 1 / f_{tap}$$

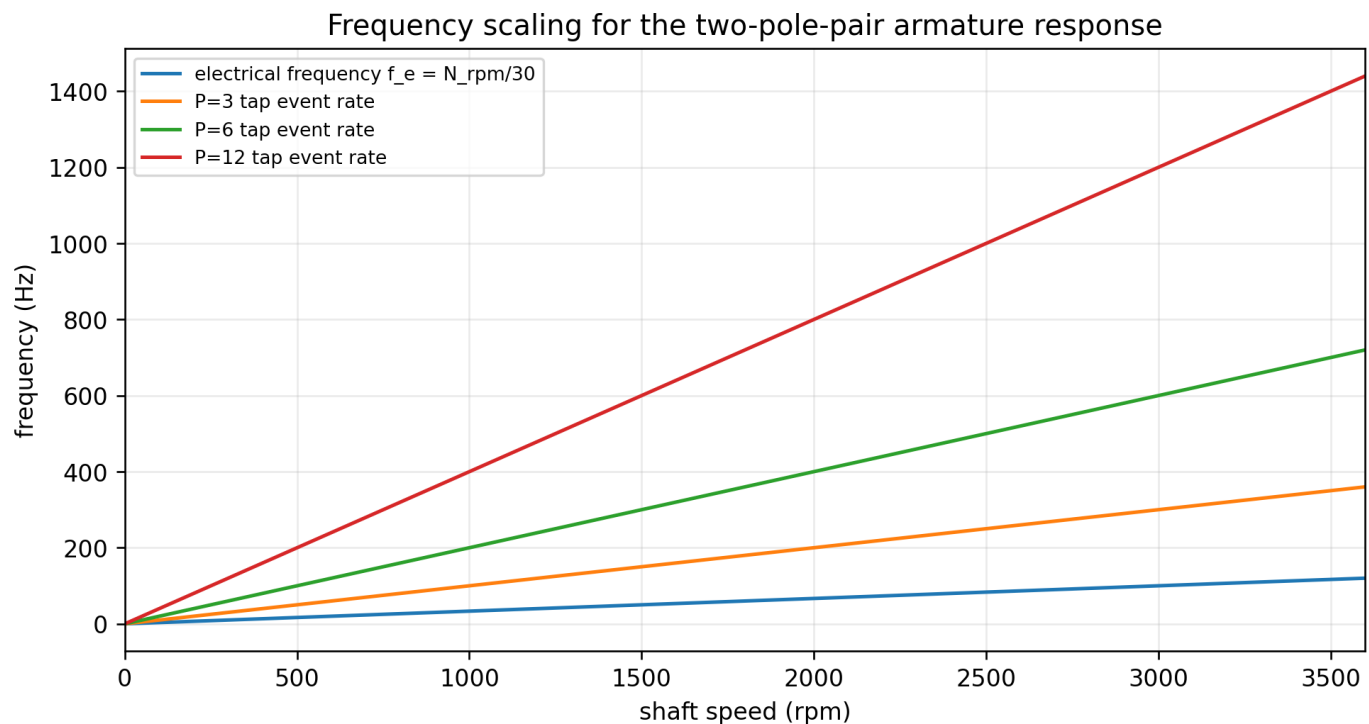


Figure 14. Electrical and tap-event frequency versus shaft speed. The electrical frequency is independent of winding-stage count P ; only the tap-event rate grows with P .

rpm	f_m Hz	f_e Hz	P=3 taps Hz	P=6 taps Hz	P=12 taps Hz
1	0.016667	0.033333	0.100000	0.200000	0.400000
10	0.166667	0.333333	1.000000	2.000000	4.000000
60	1.000000	2.000000	6.000000	12.000000	24.000000
300	5.000000	10.000000	30.000000	60.000000	120.000000
600	10.000000	20.000000	60.000000	120.000000	240.000000
1800	30.000000	60.000000	180.000000	360.000000	720.000000
3600	60.000000	120.000000	360.000000	720.000000	1440.000000

12. Polyphase Drive, Commutation, and Tap Polarity

The clean principal channel of each diametric pair is its same-polarity sum at twice shaft frequency. The opposite-polarity pair difference is the once-per-revolution index channel. The index channel can provide an absolute angular reference, while the even pair channel provides the principal rotating response.

For passive sensing or generator-style tapping, connect the pair whose signed principal channel is centered in the active sector. For active motor-style excitation, apply phase currents in the same spatial order for one rotation direction and the reverse order for the opposite direction. A response lead beta may be added to develop the desired torque angle, but beta is not fixed by scalar geometry alone.

channel or connection	effect
same-polarity 180 degree pair sum	reinforces $k=2$ principal channel and all even spatial harmonics
opposite-polarity 180 degree pair difference	isolates $k=1$ index channel and other odd spatial harmonics
90 degree mechanical offset	principal electrical opposition
equal direct sum of all P phases	cancels the balanced principal harmonic for $P>1$
staged commutation	selects successive pair maxima every $180/P$ mechanical degrees
polyphase excitation	drives all phases with $360/P$ electrical separation and a chosen lead beta

13. Physical Interpretation and Constitutive Boundary

The word pumped refers here to periodic sampling of the fixed K-Field anisotropy by mechanical rotation. The geometry determines the spatial waveform, harmonic content, phase relationships, electrical frequency ratio, diametric parity, and tap order. These are direct consequences of the supplied metric and centerline.

Absolute voltage, current, torque, mechanical work, power, and conversion efficiency require an additional constitutive law that maps cumulative scalar cost or its time derivative into an electrical or mechanical observable. The scalar cost calculation alone does not supply that conversion coefficient. Likewise, the zero-lead tap centers are geometric; an optimized motor commutation advance depends on the response dynamics and loading.

The conventional magnetic area vector of the nonplanar single coil is orthogonal to the Earth-CMB axis in the computed geometry. Conventional electromagnetic torque therefore remains a separate vector interaction from the K-Field cumulative channel. A practical device may combine both, but their contributions should be measured and modeled independently.

category	determined by present analysis
exact geometry	route, bridges, axis, Cell3 basis, nodes, and face planes
scalar transport	q_phi(s), completed cost, cumulative half-loop difference
rotational signal	H(phi), Fourier amplitudes, phases, extrema
armature phase law	two electrical cycles per mechanical revolution
pair parity	180 degree reinforcement for even channel; 90 degree principal opposition
staging law	axis spacing, phase spacing, tap sequence, event rate
not determined without device law	absolute EMF, torque, energy gain, impedance coefficient, optimal beta, efficiency

14. Reproducibility and Numerical Precision

The analysis uses the full 14,400-interval periodic centerline. Every rotation reconstructs all route points by Rodrigues rotation about the normalized Earth-CMB axis, evaluates the quadratic cost density, and performs periodic trapezoidal quadrature. Fourier coefficients are fitted over 7,200 uniformly spaced mechanical angles.

numerical quantity	value
route intervals	14400
route spacing	0.000432154216039990 R
angular sweep samples	7200
angular sweep spacing	0.05 deg
principal k=2 amplitude	2.706182128079802
principal phase	0.065460604098718 deg
Fourier fit RMS through k=20	6.915e-07
Fourier fit max error through k=20	1.084e-06
closed-loop cost rotation range	0.418134173838521
source route SHA-256	c59612142e6f9aa2925ad4b67a367bb055812e0fb82a7485ab14b24a4d514c51
coil timeline SHA-256	3b02600511a74b344a89449fd15e0a336d43c80da49cf00279b9fa861195a669

15. Foundational Consequences

consequence	meaning
A scalar cumulative state can define an armature phase.	Rotation changes the ordered half-loop accumulation even when each completed loop remains reciprocal under circulation reversal.
The dominant response is a spatial harmonic, not a guessed motor pole count.	The computed $k=2$ coefficient establishes two electrical cycles per mechanical revolution.
Diametric geometry and electrical opposition are distinct.	A 180 degree mate is in phase for $k=2$; 90 degrees is the principal opposing orientation.
Pair parity separates channels.	Even harmonics form the reinforced drive/sense channel; odd harmonics form the index channel.
Equidistant windings form a balanced phase set.	Direct summation cancels the fundamental; staged commutation or polyphase processing extracts it.
The sequence is reversible.	Changing the shaft direction reverses the tap order while preserving the same phase-spacing law.

This construction supplies a precise bridge from a fixed anisotropic scalar geometry to a rotating multiphase architecture. It converts the cumulative hysteresis of one closed route into a phase-addressable ensemble with explicit shaft-frequency scaling and a deterministic commutation order.

Appendix A. Complete Cell3 Half-Step Node Registry

Every node is $x=0.5(i a+j b+k c)$, with i,j,k in $\{-1,0,1\}$. Coordinates are native ICRS Cell3 units and are not route-sphere coordinates.

i	j	k	lambda	ICRS x native	ICRS y native	ICRS z native
-1	-1	-1	-3	-0.01097490249996122	0.00738598312742294	0.03966191101193925
-1	-1	0	-2	-0.01129009224263490	-0.00474287788114100	0.01819973372639234
-1	-1	1	-1	-0.01160528198530859	-0.01687173888970493	-0.00326244355915457
-1	0	-1	-2	-0.00987552951579954	0.01383706825676921	0.02262866024391139
-1	0	0	-1	-0.01019071925847322	0.00170820724820527	0.00116648295836448
-1	0	1	0	-0.01050590900114691	-0.01042065376035866	-0.02029569432718243
-1	1	-1	-1	-0.00877615653163786	0.02028815338611548	0.00559540947588353
-1	1	0	0	-0.00909134627431155	0.00815929237755154	-0.01586676780966337
-1	1	1	1	-0.00940653601698523	-0.00396956863101239	-0.03732894509521029
0	-1	-1	-2	-0.00078418324148799	0.00567777587921766	0.03849542805357477
0	-1	0	-1	-0.00109937298416168	-0.00645108512934627	0.01703325076802786
0	-1	1	0	-0.00141456272683536	-0.01857994613791020	-0.00442892651751905
0	0	-1	-1	0.00031518974267369	0.01212886100856393	0.02146217728554691
0	0	0	0	0.00000000000000000	0.00000000000000000	0.00000000000000000
0	0	1	1	-0.00031518974267369	-0.01212886100856393	-0.02146217728554691
0	1	-1	0	0.00141456272683536	0.01857994613791020	0.00442892651751905
0	1	0	1	0.00109937298416168	0.00645108512934627	-0.01703325076802786
0	1	1	2	0.00078418324148799	-0.00567777587921766	-0.03849542805357477
1	-1	-1	-1	0.00940653601698523	0.00396956863101239	0.03732894509521029
1	-1	0	0	0.00909134627431155	-0.00815929237755154	0.01586676780966337
1	-1	1	1	0.00877615653163786	-0.02028815338611548	-0.00559540947588353
1	0	-1	0	0.01050590900114691	0.01042065376035866	0.02029569432718243
1	0	0	1	0.01019071925847322	-0.00170820724820527	-0.00116648295836448
1	0	1	2	0.00987552951579954	-0.01383706825676921	-0.02262866024391139
1	1	-1	1	0.01160528198530859	0.01687173888970493	0.00326244355915457
1	1	0	2	0.01129009224263490	0.00474287788114100	-0.01819973372639234
1	1	1	3	0.01097490249996122	-0.00738598312742294	-0.03966191101193925

Appendix B. Cell3 Vertices and Six Support Planes

coefficient a	coefficient b	coefficient c	ICRS x	ICRS y	ICRS z
-0.5	-0.5	-0.5	-0.01097490249996122	0.00738598312742294	0.03966191101193925
-0.5	-0.5	0.5	-0.01160528198530859	-0.01687173888970493	-0.00326244355915457
-0.5	0.5	-0.5	-0.00877615653163786	0.02028815338611548	0.00559540947588353
-0.5	0.5	0.5	-0.00940653601698523	-0.00396956863101239	-0.03732894509521029
0.5	-0.5	-0.5	0.00940653601698523	0.00396956863101239	0.03732894509521029
0.5	-0.5	0.5	0.00877615653163786	-0.02028815338611548	-0.00559540947588353
0.5	0.5	-0.5	0.01160528198530859	0.01687173888970493	0.00326244355915457
0.5	0.5	0.5	0.01097490249996122	-0.00738598312742294	-0.03966191101193925

family	unit normal ICRS	negative support	positive support	separation
A_ab	[0.1940922008077677, 0.9131771688212287, 0.3583792347908041]	-0.01882857350019305	0.01882857350019305	0.03765714700038611
A_ac	[0.089052594101605, 0.8665767692910322, -0.4910339483262758]	-0.01405216690339142	0.01405216690339142	0.02810433380678284
A_bc	[-0.9959651903089612, 0.0836021625073767, -0.0326192905643998]	-0.01025436161905391	0.01025436161905391	0.02050872323810781

Appendix C. Rotational Fourier Coefficients

k	cos coefficient	sin coefficient	amplitude	phase in A cos(k phi + phase)	relative to k=2
1	7.464811243406255e-03	-1.271320739808380e-02	1.474276260564407e-02	59.5798315380 deg	5.447808724e-03
2	2.706180361872274e+00	-3.091820721730101e-03	2.706182128079802e+00	0.0654606041 deg	1.00000000e+00
3	-2.163629050604886e-05	-1.414451563292009e-05	2.584949495352010e-05	146.8257360858 deg	9.552015988e-06
4	3.685149488136805e-05	-1.635817473664149e-03	1.636232513984473e-03	88.7094660132 deg	6.046276402e-04
5	6.203774381110517e-08	-6.892370643606555e-08	9.273165029296773e-08	48.0098445797 deg	3.426659622e-08
6	3.278523706333790e-02	1.823258499348679e-03	3.283589561523657e-02	-3.1830644591 deg	1.213366066e-02
7	-1.568852849449869e-10	-2.925317725010833e-10	3.319455234747040e-10	118.2047346433 deg	1.226619303e-10
8	1.034879488364965e-05	-1.102154592156182e-04	1.107002484444070e-04	84.6358806417 deg	4.090642950e-05
9	1.260683918548278e-12	3.211039721083090e-13	1.300934934341443e-12	-14.2897433731 deg	4.807270438e-13
10	1.634998741928566e-03	2.217071890228404e-04	1.649962109799011e-03	-7.7222540181 deg	6.097010592e-04
11	7.684327877050940e-15	-3.473198587807456e-15	8.432793330348486e-15	24.3222402200 deg	3.116121876e-15
12	1.841308233200370e-06	-1.023573997212563e-05	1.040003792236460e-05	79.8021258800 deg	3.843066516e-06

Appendix D. Tap Sequences for P=1 Through P=12

P	pair axes mod 180 deg	electrical phases deg	positive order	negative order	mechanical sector deg	tap-rate factor f _{tap} /f _m
1	0	0	0-0	0-0	180.000000000	2
2	0, 90	0, 180	0-1-0	0-1-0	90.000000000	4
3	0, 60, 120	0, 120, 240	0-2-1-0	0-1-2-0	60.000000000	6
4	0, 45, 90, 135	0, 90, 180, 270	0-3-2-1-0	0-1-2-3-0	45.000000000	8
5	0, 36, 72, 108, 144	0, 72, 144, 216, 288	0-4-3-2-1-0	0-1-2-3-4-0	36.000000000	10
6	0, 30, 60, 90, 120, 150	0, 60, 120, 180, 240, 300	0-5-4-3-2-1-0	0-1-2-3-4-5-0	30.000000000	12
7	0, 25.7143, 51.4286, 77.1429, 102.857, 128.571, 154.286	0, 51.4286, 102.857, 154.286, 205.714, 257.143, 308.571	0-6-5-4-3-2-1-0	0-1-2-3-4-5-6-0	25.714285714	14
8	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.5	0, 45, 90, 135, 180, 225, 270, 315	0-7-6-5-4-3-2-1-0	0-1-2-3-4-5-6-7-0	22.500000000	16
9	0, 20, 40, 60, 80, 100, 120, 140, 160	0, 40, 80, 120, 160, 200, 240, 280, 320	0-8-7-6-5-4-3-2-1-0	0-1-2-3-4-5-6-7-8-0	20.000000000	18
10	0, 18, 36, 54, 72, 90, 108, 126, 144, 162	0, 36, 72, 108, 144, 180, 216, 252, 288, 324	0-9-8-7-6-5-4-3-2-1-0	0-1-2-3-4-5-6-7-8-9-0	18.000000000	20
11	0, 16.3636, 32.7273, 49.0909, 65.4545, 81.8182, 98.1818, 114.545, 130.909, 147.273, 163.636	0, 32.7273, 65.4545, 98.1818, 130.909, 163.636, 196.364, 229.091, 261.818, 294.545, 327.273	0-10-9-8-7-6-5-4-3-2-1-0	0-1-2-3-4-5-6-7-8-9-10-0	16.363636364	22
12	0, 15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165	0, 30, 60, 90, 120, 150, 180, 210, 240, 270, 300, 330	0-11-10-9-8-7-6-5-4-3-2-1-0	0-1-2-3-4-5-6-7-8-9-10-11-0	15.000000000	24

Appendix E. Machine-Readable Companion

The companion file K-Field_CMB_pumped_armature_analysis_1784661365.json contains the complete Cell3 basis, vertices, 27-node registry, face support planes, route centerline, fixed hysteresis metrics, 7,200-point rotational sweep, Fourier coefficients through k=20, diametric pair channels, general tap sequences for P=1 through P=12, frequency examples, source hashes, and figure hashes.

JSON section	contents
cell3_native_ICRS	basis, lengths, face areas, normals, six support planes, eight vertices, all 27 nodes
closed_route.centerline_periodic_samples	all 14,400 positions, tangents, segments, base densities, and interval costs
rotating_armature.angular_sweep	mechanical angle, single-coil signal, mate signal, pair sum, pair difference, total loop cost
rotating_armature.*_fourier	cosine and sine coefficients, amplitude, phase, residual through k=20
staged_ensemble.tap_sequences_P_1_to_12	axis angles, electrical phases, positive and negative orders, sector width, event rate
figure_manifest	file name and SHA-256 for every data-generated figure

Archival Colophon

Field	Entry
Document title	K-FIELD: CMB-PUMPED ARMATURE ANALYSIS
Project title	K-Field closed Earth-CMB route rotating armature and staged winding analysis
Principal Investigator	P. Dwayne Esterline
Credentials	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx
Output PDF	K-Field_CMB_pumped_armature_analysis_1784661365.pdf
Machine-readable companion	K-Field_CMB_pumped_armature_analysis_1784661365.json
Generation epoch	1784661365
Page size	US Letter
Coordinate frame	ICRS Cartesian; route sphere R=1; Cell3 retained in separate native units
Fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Source route	K-Field_max_residual_closed_CMB_route_1784650136(2).json
Source route SHA-256	c59612142e6f9aa2925ad4b67a367bb055812e0fb82a7485ab14b24a4d514c51